

sTMS Interface Functional Specification

Nanyang Polytechnic – CRM System

Version 1.0

Document Version

Version	Effective Date	Summary of Changes	Created/ Updated By
0.1	22/08/2022	Initial version of NYP CRM System interface functional specification	AvePoint
0.2	14/09/2022	Revised base on NYP feedback. • Update section 1.1. This interface functional spec is part of the general requirement tender spec; • Update section 2.1. sTMS shall transfer empty files with header info; • Update section 3.1. Update remarks and field mapping; • Remove 3.2 pickup list. After sTMS send the value NYP CRM side will do the field mapping; • Add section 4.2. NYP CRM will send a notification email describing the interface job details. (Check Sum)	AvePoint
0.3	07/10/2022	Revised base on NYP feedback. • Update section 1.1 description; • Add CheckSum in 2.1; • Update 3.1 and change 'Residential Address' to 'Mailing Address'; • Add email notification template in 5. Appendix	AvePoint
0.4	13/10/2022	Revised base on NYP feedback. • 2.1 Delete the email notification. • 2.1 & 3.2 Add checksum and sample file • Remove 4. Assumption and Constraints and 5. Appendix (Checksum send txt file instead of email notification)	AvePoint
1.0	18/10/2022	Official version for acceptance and sign-off	AvePoint

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1. Introduction

1.1 Purpose of this document

This Interface Functional Specification (IFS) is to provide the agreed baseline of communication requirements and detailed technical specification between the SaaS shared Training Management System (sTMS) and NYP CRM System. This interface will be a 2-way interface between the system, sTMS required to transfer CET graduate contact, employment history, course, qualification, award information to NYP CRM System. When applicants apply for a course which eligible for alumni discount in sTMS, sTMS shall call NYP CRM System API with NRIC and birthday to check the alumni status.

The purpose of this interface is for NYP alumni to enjoy alumni discounts when applying for specific courses in sTMS. It is required to push sTMS CET graduate information to NYP CRM System and NYP CRM System is the source of truth for sTMS to check the CET and PET alumni status.

This NYP CRM interface constitutes part of the general requirement tender specification (Part 2, Section B – Functional Requirements – Section 15. Alumni Management Interface) and all payment(s) related to this interface has already been catered for under the existing main payment schedule.

1.2 Intended Audience

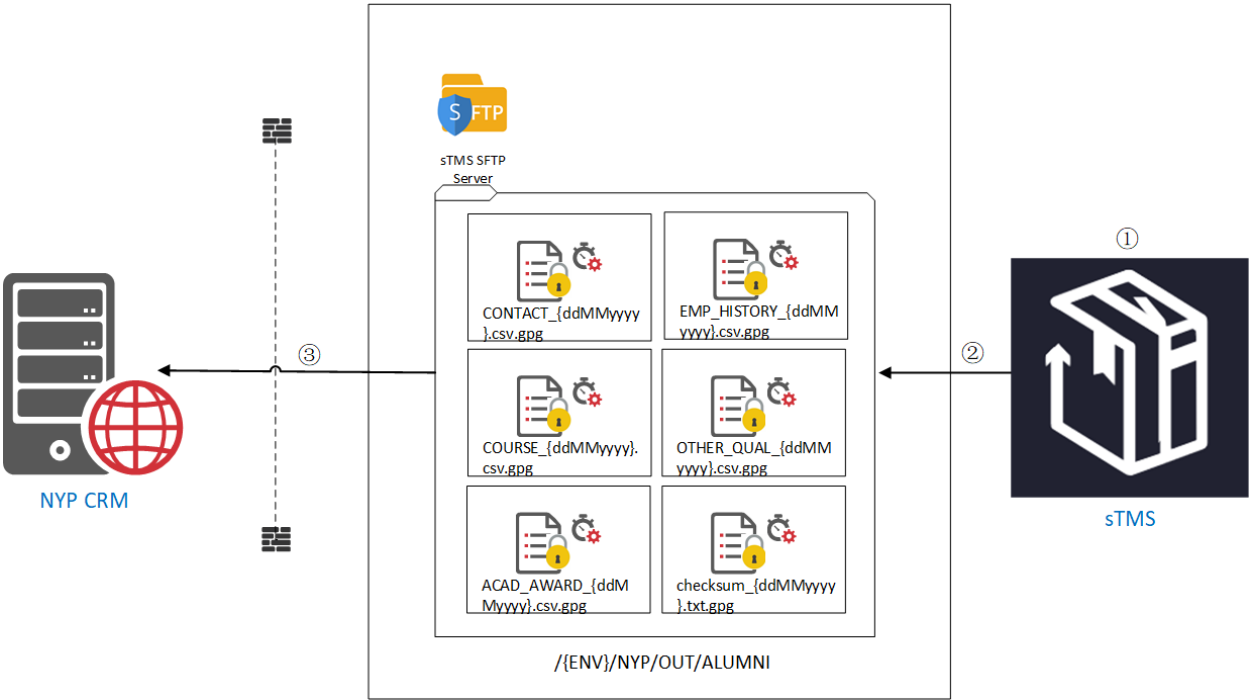
This technical document is used for system-to-system communication. Therefore, the intended audiences are the technical stakeholders of sTMS and NYP CRM System.

2. Business Overview

2.1 From sTMS to NYP CRM System

There are 3 roles involved in the interface diagram.

- **sTMS**: shared Training Management System.
- **sTMS SFTP Server**: A SFTP server hosted in Azure that stores encrypted files used for data exchange.
- **NYP CRM System**: A NYP internal system that contain NYP alumni details.



SN	Description
1	sTMS generates CSV files of NYP CET graduate info on a pre-defined schedule base on system setting or manually triggered by CET users (File generation rule please refer to 2.3 Business Rule).
2	sTMS configures timer job to push CSV files into sTMS SFTP server after encryption with PGP protocol.
3	NYP CRM System pulls the NYP CET graduate student information from sTMS SFTP server on demand.

Output Interface file

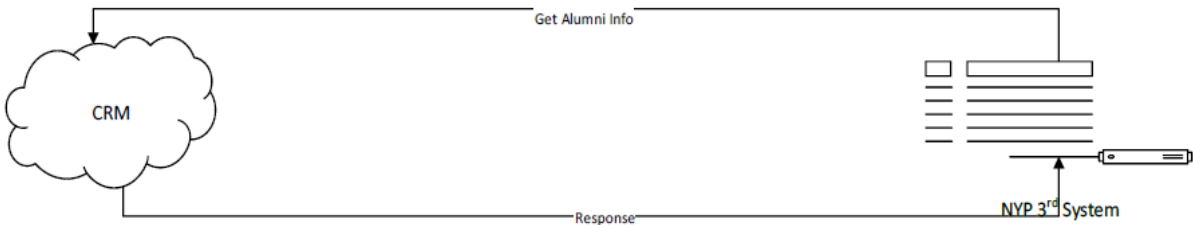
Interface File Name	CONTACT_{ddMMyyyy}.csv.gpg EMP_HISTORY_{ddMMyyyy}.csv.gpg COURSE_{ddMMyyyy}.csv.gpg OTHER_QUAL_{ddMMyyyy}.csv.gpg ACAD_AWARD_{ddMMyyyy}.csv.gpg checksum_{ddMMyyyy}.txt.gpg
Data Source System	shared Training Management System
Data Destination System	Nanyang Polytechnic CRM System
Method	SFTP
File Path	/{ENV}/NYP/OUT/ALUMNI ENV: UAT, PROD
Encryption Method	PGP
Data Syncing Method	Incremental synchronization. Graduate information will only be sent once for students under the same course code. If the student graduate from another course that allows him to become NYP alumni again, sTMS will send all relevant graduate information to NYP.
Frequency	Daily
Timing	The cut off time for data will be 23:59:59 pm every day, timer job trigger timing is next day 01:00 am (Configurable)
Folder Retention Policy	Files under the /{ENV}/NYP/OUT/ALUMNI are retained for a maximum of 30 days in the sTMS SFTP server, after this period the files will be automatically deleted. NYP CRM team will provide the script file to housekeep the files and can delete these files after pulling out.
File Transfer Policy	sTMS shall transfer empty files with header info into sTMS SFTP server if there are no new records are added/updated in the table.
Possible Exception and Exception Handling	File loss or transfer failure due to sTMS internal exception. E.g., Timer service down, etc. Exception handling steps: • sTMS support team shall be informed by email of the transfer failure. • sTMS support team checks the relevant logs to troubleshoot the cause of the failure. • If the issue is due to the transfer file could not being found in sTMS SFTP server: ◦ sTMS support team redeposits the file to sTMS SFTP server. The CSV file generated by sTMS have records missing. Exception handling steps:

	• NYP team raise a ticket to sTMS support team for investigation. • sTMS support team checks the relevant logs to troubleshoot the cause of the failure. • sTMS support team redeposits the file to sTMS SFTP server.
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2.2 From NYP CRM System to sTMS

There are 2 roles involved in the interface diagram.

- **NYP 3rd System:** shared Training Management System.
- **NYP CRM System:** A NYP internal system that contain NYP alumni details.



Interface Name	Check Alumni Status
Description	AvePoint allow NYP 3rd system check alumni status by this interface
Hosting System	NYP CRM System
Data Source System	NYP CRM System
Data Destination System	Shared Training Management System (sTMS)
Mode of Interface	API
Method	POST
Content-Type	application/json
Frequency	Ad-hoc, please refer to 2.3 Business Rule
API URL	https://[DynamicsUrl]/api/data/v9.2/ave_CheckAlumniStatus
Request Body	{ "ave_contactalumninrictext":"NRIC", // last 5 characters "birthdate": "birth date" // format is ddMMyyyy }

Return Value	<pre>{ "Alumni": "Yes/No", "ErrorCode": "Only has value when data validation error and http status code is equal to 200", "ErrorMessage": "Only has value when data validation error and http status code is equal to 200" }</pre>
Response Code	200 or other http status code

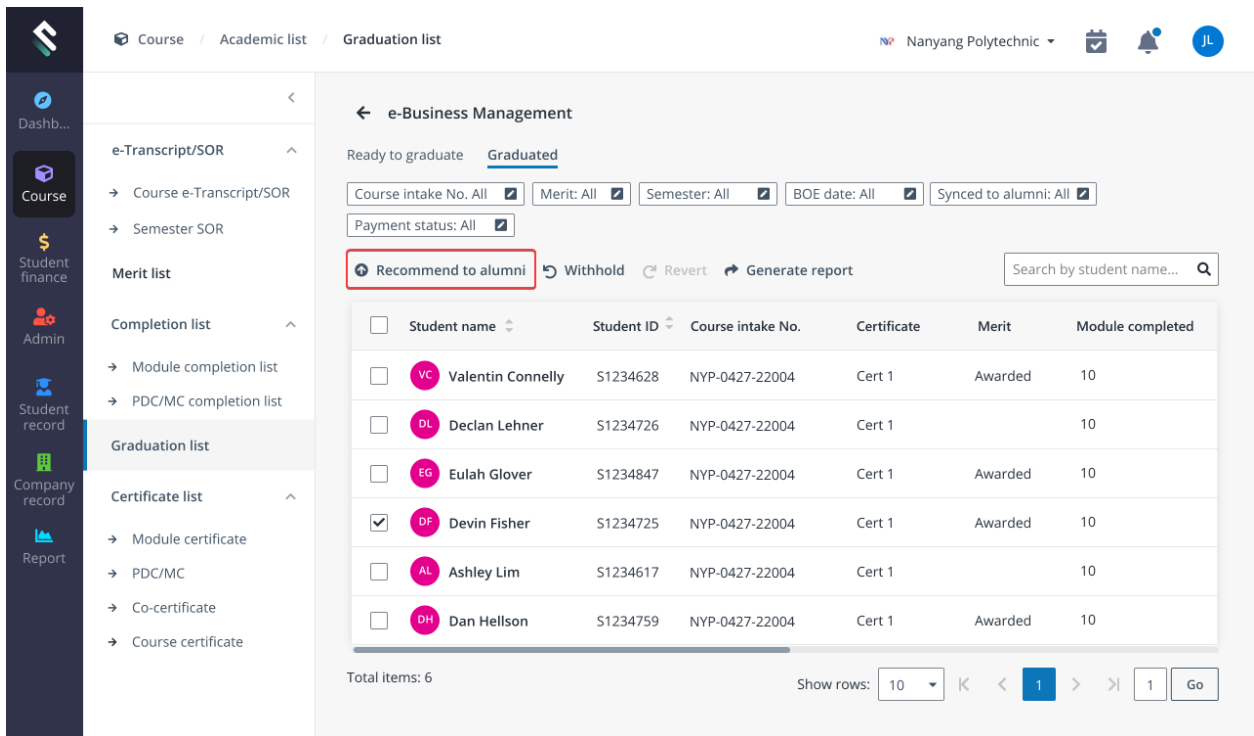
2.3 Business Rule

sTMS has automatic and manual configuration methods to send NYP alumni info to NYP Alumni System. If one method is successful, sTMS will not send repeated alumni info to NYP again.

2.3.1 Manually Add Eligible Graduate Records to NYP CRM System

The build-in configuration is set as a 'manual' action named "Recommend to alumni" in sTMS to disseminate course completion details to NYP Alumni System. By performing this action, it will trigger the system flow to send the graduated list information from sTMS to NYP Alumni System. For graduation list details please refer to sTMS stage 3 functional specification: 7.4.6 Graduation List.

Refer to the clickable prototype here: [Graduation List](#)



Course / Academic list / Graduation list

Nanyang Polytechnic

e-Business Management

Ready to graduate **Graduated**

Course intake No.: All Merit: All Semester: All BOE date: All Synced to alumni: All

Payment status: All

Recommend to alumni Withhold Revert Generate report

Search by student name...

☐	Student name	Student ID	Course intake No.	Certificate	Merit	Module completed
☐	VC Valentin Connelly	S1234628	NYP-0427-22004	Cert 1	Awarded	10
☐	DL Declan Lehner	S1234726	NYP-0427-22004	Cert 1		10
☐	EG Eulah Glover	S1234847	NYP-0427-22004	Cert 1	Awarded	10
☒	DF Devin Fisher	S1234725	NYP-0427-22004	Cert 1	Awarded	10
☐	AL Ashley Lim	S1234617	NYP-0427-22004	Cert 1		10
☐	DH Dan Hellson	S1234759	NYP-0427-22004	Cert 1	Awarded	10

Total items: 6

Show rows: 10

1 1 Go

2.3.2 Automatically Add Eligible Graduate Records to NYP CRM System

The system also allows NYP to configure automated flows on top of the build-in manual mode, the configuration is also flexible down to the specific course level if such a configuration is required. For alumni configuration details please refer to sTMS stage 3 functional specification: 11. Alumni Management Interface.

Refer to the clickable prototype here: [Alumni Setting](#)

Admin

Service management

Alumni setting

Role management

Account management

Master data

Form bank

Rule management

Service management

Email notification

Template

Document template

Digital certificate template

Certificate template

Transcript template

Signatures

Certificate image library

Admin settings

Announcement management

Student resource

Alumni management

Finance settings

Odd-cents write-off setting

Refund priority setting

Bank file setting

Student virtual card

User activity report

Status: All

Create

Edit

Activate

Deactivate

Name

Alumni Setting of Course type

Total items: 1

Create alumni setting

Name

Description

Scope configuration

Select the scope to apply

A combination of factors

Specific courses

Course category

Part-time Diploma (PTD)

Certificate course

Course type

Academic Full Qualification (FQ)

SkillFuture Series

Branch

SIRS

School

School of business

Details

Choose a condition

On course completion date

10

Weeks

after course completion date

Finish

% modules and are competent

on result release date

10

Weeks

after result release date

upon graduation

Cancel

Save

2.3.3 Check Alumni Status and Generate Billing

sTMS allows NYP to configure if alumni discount is enabled during the course setup. If enabled, the student who is in Alumni will get the corresponding discount setup by the IHL. For course setup details, please refer to sTMS stage 1 functional specification: 6.3 Manage course. For discount setup details, please refer to sTMS stage 1 functional specification: 5.8 Manage Discount.

Course

Student finance

Admin

Version 2

<

Course configuration

1 General information

2 Course details

3 Course fee

Course / Course configuration / Create course

4

Course fee

Preview

Course setup

Course type *
Academic Certificate

Billing by *
PDC/MC

Rate effective date base on *
Course intake date

Allow payment by *
PSEA x SFC x Credit/Debit Card x

Course rate set up by
Course rate table

Fee setup

Course
Specialist diploma in e-Business Managem...

Post Diploma Certificate
Post Diploma in MSc e-Business Manage...

Module (e-Business Management)
Statement of result

Module (Digital Development with ...
None

Post Diploma Certificate
Specialist diploma in Digital Developme...

Module (Digital Development with ...
Statement of result
Statement of participation

Course rate setup

Rate code
PTD-AC-BMO-NWSP

Rate version
Version 1

Rate type
Hourly rate per pax

Current rate
21.8

Funding
☒ SSG Cat A View details

Supplementary fee
☒ Miscellaneous fee View details

Discount
☒ Alumni discount View details
☐ General discount View details

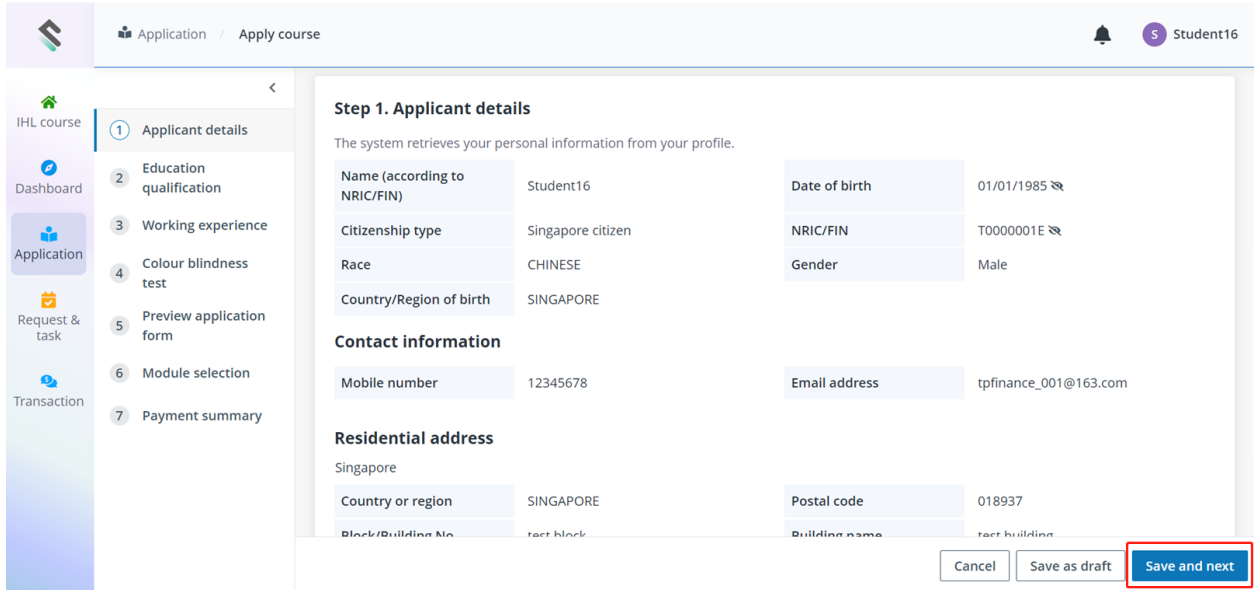
Cancel

Save as draft

Back

Complete

When a student applies for course intake that is eligible for the alumni discount, after the student complete Step 1 Application details and click "Save and next" button, the system will call the Check Alumni Status API to check if the student is an NYP alumni. If yes, the student will get the corresponding discount setup by NYP.



Step 1. Applicant details

The system retrieves your personal information from your profile.

Name (according to NRIC/FIN)	Student16	Date of birth	01/01/1985
Citizenship type	Singapore citizen	NRIC/FIN	T0000001E
Race	CHINESE	Gender	Male
Country/Region of birth	SINGAPORE		

Contact information

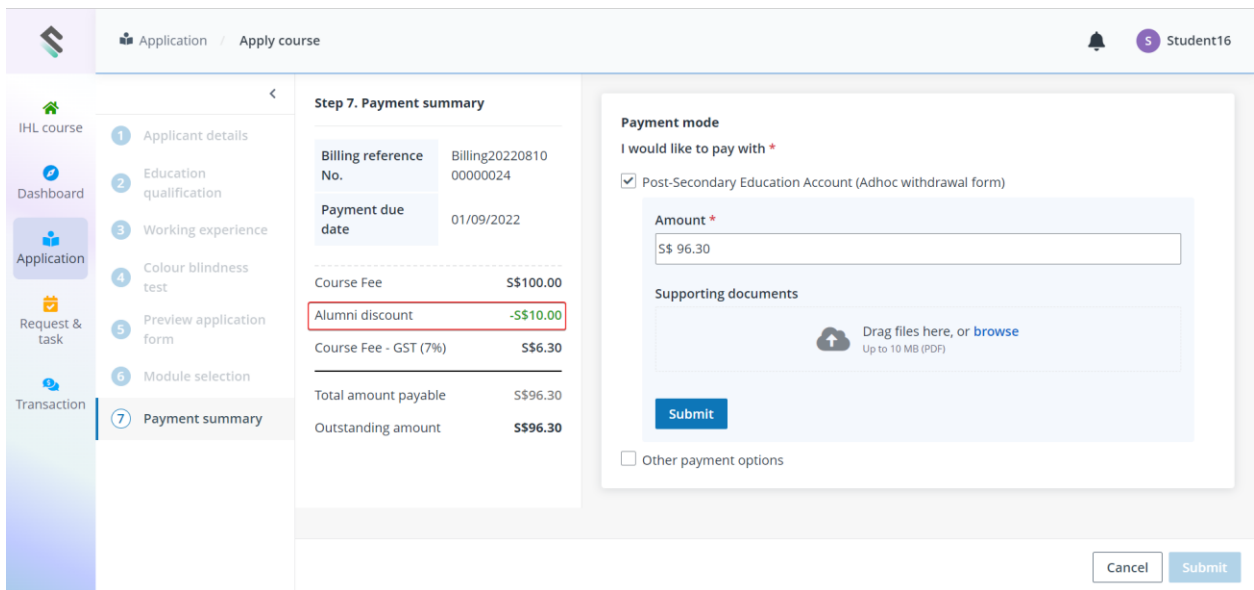
Mobile number	12345678	Email address	tpfinance_001@163.com
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Residential address

Singapore

Country or region	SINGAPORE	Postal code	018937
Block/Building No.	test block	Building name	test building

Buttons: Cancel, Save as draft, **Save and next**



Step 7. Payment summary

Billing reference No.	Billing20220810 00000024
Payment due date	01/09/2022

Course Fee	S\$100.00
Alumni discount	-S\$10.00
Course Fee - GST (7%)	S\$6.30
Total amount payable	S\$96.30
Outstanding amount	S\$96.30

Payment mode

I would like to pay with *

☒ Post-Secondary Education Account (Adhoc withdrawal form)

Amount *

S\$ 96.30

Supporting documents

Drag files here, or [browse](#)
Up to 10 MB (PDF)

Submit

☐ Other payment options

Buttons: Cancel, **Submit**

3 Interface Fields

3.1 CSV Files Interface Field

CONTACT

CRM Field Name	sTMS Field Name	Description	Data Type	Remarks
NRIC or Passport Number	NRIC/FIN/Passport No.	Alumni's NRIC/Passport number/FIN	String(\$nvarchar(15))	Send Alumni's NRIC/FIN number. If sTMS don't have NRIC/FIN, send passport number
Last Name	Name	Alumni's name	String(\$nvarchar(100))	
Alumni Address 1	Mailing Address ([Block Number], [Street Name])	Alumni's home address	String(\$nvarchar(30))	Sample Value: BLK 133, Street A
Alumni Address 2	#[Unit Level]-[Unit Number], [Building Name]	Alumni's home address	String(\$nvarchar(30))	Sample Value: #07-21, Building A
Alumni Address 3	-	-	String(\$nvarchar(15))	Default to Null
Postal Code	Postal Code	Alumni's home postal code	String(\$nvarchar(6))	
Country	Country code	Alumni's home country	String(\$nvarchar(50))	
Mobile Number	Mobile number	Alumni's personal mobile number	String(\$nvarchar(30))	
Personal Email	Email	Alumni's personal email address	String(\$nvarchar(100))	
Gender	Gender	Alumni's gender	String(\$nvarchar(7))	
Race	Race	Alumni's race	String(\$nvarchar(200))	
Birthday	Date of birth	Alumni's date of birth	String(\$ddMMyyyy))	Sample value: 11022022

Citizenship	Nationality	Alumni's citizenship	String(\$nvarchar(200))	
PR Status	Citizenship type	Alumni's SG PR status	String(\$nvarchar(10))	
Date Joined	Date from	Date of joining company	String(\$ddMMyyyy))	Sample value: 02032021 Have value when working experience end date select "Current date"
Designation	Position held	Alumni's current job designation	String(\$nvarchar(200))	Have value when working experience end date select "Current date"
Job Function	Field of Work	Alumni's current job function	String(\$nvarchar(200))	Sample value: IT & Other Information Svcs Have value when working experience end date select "Current date"
Company	Name of employer	Alumni's current company name	String(\$nvarchar(200))	Have value when working experience end date select "Current date"
NS Release Date	Complete NS. ORD date	Date of male alumni's NS release	String(\$ddMMyyyy))	Sample value: 02032020

EMP_HISTORY

CRM Field Name	sTMS Field Name	Description	Data Type	Remarks
NRIC or Passport Number	NRIC/FIN/Passport No.	Alumni's NRIC/Passport number/FIN	String(\$nvarchar(15))	Send Alumni's NRIC/FIN number. If sTMS don't have NRIC/FIN, send passport number
Date Joined	Date from	Date of joining company	String(\$ddMMyyyy))	Sample value: 02032021
Designation	Position held	Alumni's former job designation	String(\$nvarchar(200))	
Job Function	Field of Work	Alumni's former job function	String(\$nvarchar(200))	Sample value: IT & Other Information Svcs
Company	Name of employer	Alumni's former company name	String(\$nvarchar(200))	

COURSE

CRM Field Name	sTMS Field Name	Description	Data Type	Remarks
NRIC or Passport Number	NRIC/FIN/Passport No.	Alumni's NRIC/Passport number/FIN	String(\$nvarchar(15))	Send Alumni's NRIC/FIN number. If sTMS don't have NRIC/FIN, send passport number
Admission number	Student ID	Admission Number	String(\$nvarchar(15))	UAM Common ID
Campus code	Branch code	Campus	String(\$nvarchar(12))	
School code	School	School Code	String(\$nvarchar(12))	
Admission date	Enrolled Date	Admission Date	String(\$ddMMyyyy)	Sample value: 03032022
Graduation date	Graduation date	Graduation Date	String(\$ddMMyyyy)	Sample value: 03032023
Course code	Course Code	Course Code	String(\$nvarchar(20))	Sample value: NYP-SD-FQ0001
Course Name	Course Name	Course Name	String(\$nvarchar(200))	
Date of completion	Certificate issue date	Date student conferred with diploma	String(\$ddMMyyyy)	Sample value: 03012023

OTHER_QUAL

CRM Field Name	sTMS Field Name	Description	Data Type	Remarks
NRIC or Passport Number	NRIC/FIN/Passport No.	Alumni's NRIC/Passport number/FIN	String(\$nvarchar(15))	Send Alumni's NRIC/FIN number. If sTMS don't have NRIC/FIN, send passport number
Name of Institution	Award institution	Name of institution	String(\$nvarchar(20))	

Other Qualification	Exam/Cert	Name of other qualification attained	String(\$nvarchar(100))	
Year of Graduation	Graduation date	Year of award	String(\$nvarchar(4))	Sample value: 2008
Full-Time/Part-Time	Full time/Part time	Part time or full time study (P/F)	String(\$nvarchar(1))	Value list: P – Part-time, F – Full-time
Qualification Type	Qualification type	Qualification type	String(\$nvarchar(100))	Sample value: Secondary

ACAD_AWARD

CRM Field Name	sTMS Field Name	Description	Data Type	Remarks
NRIC or Passport Number	NRIC/FIN/Passport No.	Alumni's NRIC/Passport number/FIN	String(\$nvarchar(15))	Send Alumni's NRIC/FIN number. If sTMS don't have NRIC/FIN, send passport number
Year of Award	Disbursement date	Year of award	String(\$nvarchar(4))	Sample value: 2008
Award	Award name	Name of academic award/prize	String(\$nvarchar(100))	Application type is Award
Institution	IHL	Name of awarding institute/university	String(\$nvarchar(4))	Default to NYP
Scholarship	Award name	Name of scholarship	String(\$nvarchar(100))	Application type is scholarship

3.2 Sample Files

					
CONTACT_25082022.csv	EMP_HISTORY_25082022.csv	COURSE_22082022.v	OTHER_QUAL_25082022.csv	ACAD_AWARD_25082022.csv	checksum_25082022.txt